

Week 7: Modeling Predictable Returns

Motivation

1. The goal of investing is to earn a profit.
2. The most meaningful measure of profit for investors is returns.
3. As we've seen, there is mixed evidence for the EMH, i.e., seems to be satisfied for some stocks but not others $\implies$ validity of the EMH is questionable.
4. When EMH is not satisfied, then we can try to predict, or forecast, returns.

Returns, Prices, without the EMH

- ▶ When there is evidence to suggest EMH is not satisfied...
- ▶ We want to model the dynamics of returns and use them to generate forecast for returns and for prices.
- ▶ The most common (and arguably the simplest) approaches to forecast are autoregressive (AR) models.
- ▶ First, we need to *verify* that returns are stationary.

Revision of properties of expectations

There are some simple properties of expectations. If y is a random variable and a is a constant then

$$E(a) = a \qquad \text{var}(a) = 0$$

$$E(ay) = aE(y) \qquad \text{var}(ay) = a^2 \text{var}(y)$$

$$E(a + y) = a + E(y) \qquad \text{var}(a + y) = \text{var}(y)$$

Revision of properties of expectations

If y_1 and y_2 are random variables then

$$E(y_1 + y_2) = E(y_1) + E(y_2)$$

and

$$\text{var}(y_1 + y_2) = \text{var}(y_1) + \text{var}(y_2) + 2 \text{cov}(y_1, y_2).$$

Stationary

A time series Y_t is defined to be covariance *stationary* if

1. $E(Y_t) = \mu$ does not depend on t
2. $\text{var}(Y_t) = \sigma^2$ does not depend on t
3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t , for $j = 1, 2, \dots$

Random Walk - Non-stationary

Random Walk Model

1. A useful model of log stock prices, p_t , is the **random walk model**.
2. Let ϵ_t be white noise with mean 0 and variance σ^2 .
3. Let p_0 be an arbitrary starting point.

Random Walk Model

The variable p_t is called a **random walk** with increments ϵ_t if:

$$\begin{aligned} p_t &= p_{t-1} + \epsilon_t, \quad t \geq 1 \\ &= p_0 + \epsilon_t + \cdots + \epsilon_1. \end{aligned}$$

Assuming p_0 is fixed and for simplicity $p_0 = 0$.

1. The mean is

$$\mathbb{E}[p_t] = 0$$

because

$$\mathbb{E}(p_t) = \mathbb{E}(\epsilon_t + \dots + \epsilon_1) = \mathbb{E}(\epsilon_t) + \dots + \mathbb{E}(\epsilon_1) = 0$$

2. The variance is

$$\mathbb{V}[p_t] = \sigma^2 \cdot t$$

because

$$\begin{aligned}\mathbb{V}(p_t) = \mathbb{V}(\epsilon_t + \dots + \epsilon_1) &= \mathbb{V}(\epsilon_t) + \dots + \mathbb{V}(\epsilon_1) \\ &= \sigma^2 + \dots + \sigma^2 \\ &= \sigma^2 t\end{aligned}$$

Random Walk Model with drift

Log stock prices, and more so market indices, tend to have a small and positive mean. Therefore, the random walk model becomes a random walk with drift:

$$\begin{aligned} p_t &= \mu + p_{t-1} + \epsilon_t, \quad t \geq 1 \\ &= \mu \cdot t + p_0 + \epsilon_t + \cdots + \epsilon_1. \end{aligned}$$

Random walk with drift

Assuming p_0 is fixed and for simplicity $p_0 = 0$.

The mean is

$$\mathbb{E}[p_t] = \mu \cdot t$$

The variance is

$$\mathbb{V}[p_t] = \sigma^2 \cdot t$$

To test whether the log price of an asset follows a random walk or a random walk with drift, we employ the models:

$$p_t = \phi_1 p_{t-1} + \epsilon_t,$$

or

$$p_t = c + \phi_1 p_{t-1} + \epsilon_t.$$

This would require us to test:

$$H_0 : \phi_1 = 1$$

$$H_1 : \phi_1 < 1$$

Let $\Delta y_t = y_t - y_{t-1}$

More generally, for a given time series y_t , we can test for the existence of a unit root using the regression

$$y_t = c_t + \phi_1 y_{t-1} + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \epsilon_t$$

Here y_t is the series in **levels**.

The idea of the additional lagged differences $\Delta y_{t-1}, \dots, \Delta y_{t-p}$ is to model the autocorrelation so that the residuals of the equation are white noise. The best way to choose this lag length p is using BIC.

The hypotheses are

$$H_0 : \phi_1 = 1 \quad \text{versus} \quad H_1 : \phi_1 < 1.$$

A convenient way to test $H_0 : \phi_1 = 1$ is to subtract y_{t-1} from both sides, giving

$$\Delta y_t = c_t + \phi_{c1}y_{t-1} + \sum_{i=1}^{p-1} \beta_i \Delta y_{t-i} + \epsilon_t,$$

where

$$\phi_{c1} = \phi_1 - 1$$

Here, Δy_t is the series in first differences.

Equivalently, the hypotheses can be written as

$$H_0 : \phi_{c1} = 0 \quad \text{versus} \quad H_1 : \phi_{c1} < 0.$$

Thus, this is a left-tailed test.

Augmented Dickey-Fuller Unit Root Test

If y_t clearly does not have a time trend, then

$$c_t = c,$$

where c is a constant. In R, this is called `type = "drift"`.

If y_t clearly has a linear time trend, then

$$c_t = c_0 + c_1 t$$

In R, this specification is called `type = "trend"`.

Augmented Dickey-Fuller Unit Root Test

The t-ratio

$$ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})},$$

are referred to as the Augmented Dickey-Fuller (ADF) unit root tests.

The critical values for this statistic have been tabulated using simulations.

We reject the null hypothesis if the ADF test is less than the appropriate critical value.

ADF test in R

```
library(urca)

adf <- ur.df (y,
              type = "trend",      # "drift", or "trend"
              selectlags = "BIC",
              lags = 12            # max lag
              )

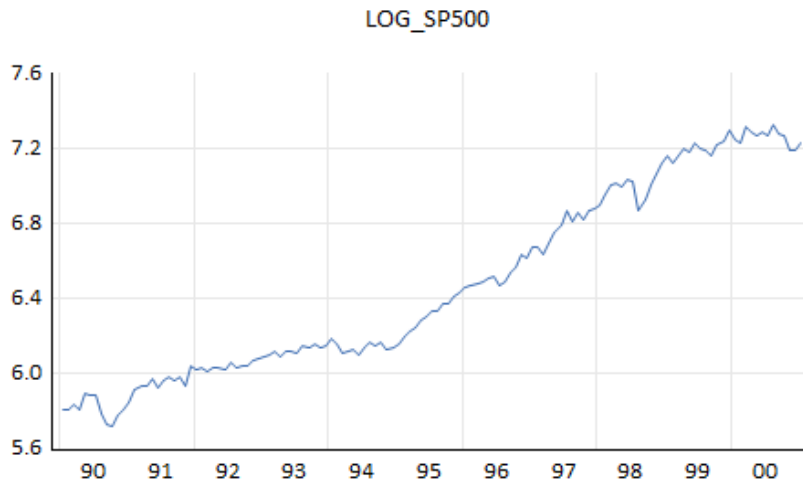
summary(adf)
```

Example: Testing for a unit root in log stock prices

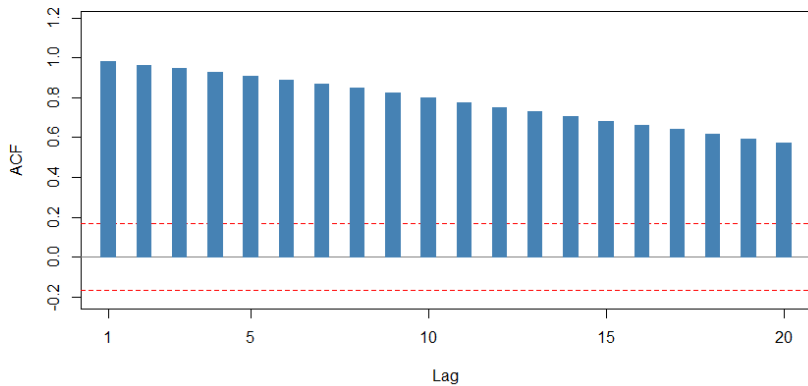
The log of asset prices is usually modelled as a random walk with drift.

Consider testing for a unit root in the log of the monthly S&P 500 index, p_t over the period January 1990 through January 2001.

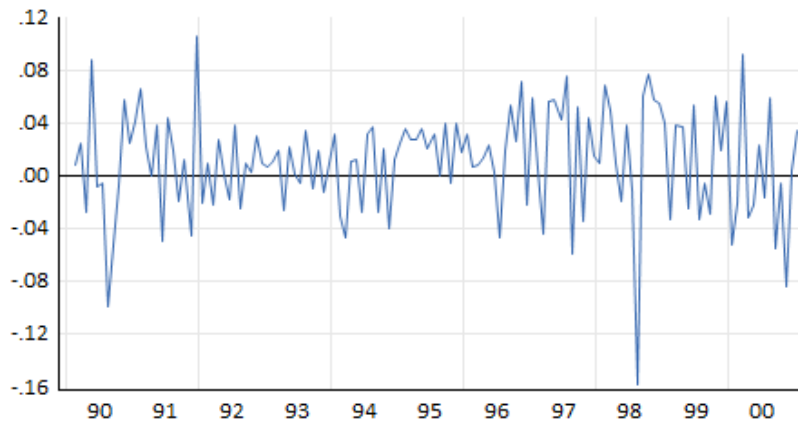
Log of S&P 500 index, Jan 1990 to Jan 2001

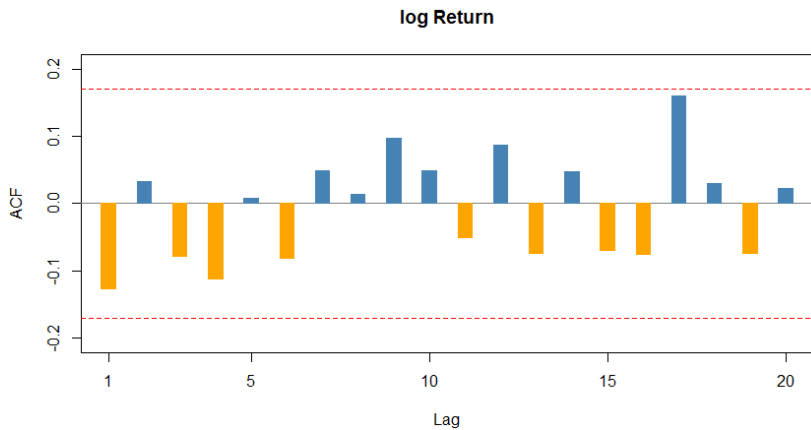


log SP500



Log Return of S&P 500 index





Let us now conduct a statistical test to determine whether or not log of S&P 500 is non-stationary. This would require us to test, in the ADF regression:

$$H_0 : \phi_{c1} = 0$$

$$H_1 : \phi_{c1} < 0$$

The test statistic is

$$ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})} = -1.652$$

The 5% critical value is $\tau_3 = -3.43$. Since

$$-1.652 > -3.43$$

we fail to reject H_0 at the 5% significance level. Therefore, there is insufficient evidence to conclude that $\log(\text{S\&P 500})$ is stationary.


```
log_sp500 = log(data.xts$sp500)

adf_level <- ur.df(log_sp500,
                   type = "trend",
                   selectlags = "BIC",
                   lags = 12)

summary(adf_level)
```

```
#####  
# Augmented Dickey-Fuller Test Unit Root Test #  
#####
```

Test regression trend

Call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.167406	-0.024106	0.000845	0.020012	0.089164

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.3191811	0.1850001	1.725	0.0871 .
z.lag.1	-0.0549851	0.0332845	-1.652	0.1012
tt	0.0007282	0.0004547	1.602	0.1120
z.diff.lag	-0.1474630	0.0919790	-1.603	0.1116

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03794 on 116 degrees of freedom
Multiple R-squared: 0.05354, Adjusted R-squared: 0.02907
F-statistic: 2.187 on 3 and 116 DF, p-value: 0.09325

value of test-statistic is: -1.652 5.5772 1.3655

critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

Let us now conduct a statistical test to determine whether or not the log returns of S&P 500 is non-stationary. Again, in the ADF regression, we test:

$$H_0 : \phi_{c1} = 0$$

$$H_1 : \phi_{c1} < 0$$

The test statistic is

$$ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})} = -8.600$$

The 5% critical value is $\tau_2 = -2.88$. Since

$$-8.600 < -2.88,$$

we reject H_0 at the 5% significance level and conclude that the log returns are stationary.

```
log_ret <- na.omit(diff(log_sp500))  
  
# we select "drift"  
adf_log.ret <- ur.df(log_ret,  
                      type = "drift",  
                      selectlags = "BIC",  
                      lags = 12)  
summary(adf_log.ret)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Test regression drift

call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.17242	-0.02565	0.00233	0.02269	0.08406

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.013446	0.003838	3.503	0.000654	***
z.lag.1	-1.211969	0.140930	-8.600	4.41e-14	***
z.diff.lag	0.021691	0.091784	0.236	0.813594	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03798 on 116 degrees of freedom

Multiple R-squared: 0.5968, Adjusted R-squared: 0.5899

F-statistic: 85.86 on 2 and 116 DF, p-value: < 2.2e-16

value of test-statistic is: -8.5998 36.9863

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

Stationary Processes

AR(1) processes

Recall that we define $\mathcal{F}_{t-1}$ to be all information available to the econometrician at time $t - 1$.

This could consist of possible lagged values of y_t , such as $y_{t-1}, y_{t-2}, \dots, y_1$.

MDS

Definition

A time series ε_t is an martingale difference sequence (MDS) if

$$E(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$$

Here $\mathcal{F}_{t-1}$ = the information set available up to time $t - 1$

$E(\varepsilon_t \mid \mathcal{F}_{t-1})$ = the conditional expectation of u_t given past information.

$E(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$ means that the expected value of the current shock, given all past information, is zero.

AR(1) processes

A time series y_t is an **AR(1) process** if

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t$$

Assume ε_t is an MDS and $\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \mathbb{V}(\varepsilon_t) = \sigma_\varepsilon^2$

Unconditional Mean and Variance

We will show that: (a) the unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1}$$

(b) the unconditional variance

$$\mathbb{V}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \phi_1^2}$$

An AR(1) process is **covariance (weakly) stationary** if $|\phi_1| < 1$.

AR(1) processes

The **conditional mean** of y_t is

$$\begin{aligned}\mathbb{E}[y_t | \mathcal{F}_{t-1}] &= \mathbb{E}[c + \phi_1 y_{t-1} + \varepsilon_t | \mathcal{F}_{t-1}] \\&= \mathbb{E}[c | \mathcal{F}_{t-1}] + \mathbb{E}[\phi_1 y_{t-1} | \mathcal{F}_{t-1}] + \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] \\&= c + \phi_1 \mathbb{E}[y_{t-1} | \mathcal{F}_{t-1}] + 0 \\&= c + \phi_1 y_{t-1}.\end{aligned}$$

AR(1) processes

- ▶ We can estimate the AR(1) model using OLS:
- ▶ Choose $(\hat{c}, \hat{\phi}_1)$ such that

$$(\hat{c}, \hat{\phi}_1) = \arg \min \sum_{t=2}^T (y_t - c - \phi_1 y_{t-1})^2.$$

- ▶ This is a regression of y_{t+1} on a constant c and the regressor y_t .
- ▶ All OLS diagnostics apply.

Question:

What is the unconditional mean of y_t ?

What is $\mathbb{E}(y_t)$?

Hint: Use LIE

Unconditional Mean

$$\mathbb{E}(y_t) = \mathbb{E}(\mathbb{E}(y_t \mid \mathcal{F}_{t-1}))$$

$$= \mathbb{E}(c + \phi_1 y_{t-1})$$

$$= c + \phi_1 \mathbb{E}(y_{t-1})$$

$$= c + \phi_1 \mathbb{E}(y_t)$$

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1}$$

Conditional variance of AR(1) processes

The **conditional variance** of y_t is

$$\begin{aligned}\mathbb{V}(y_t | \mathcal{F}_{t-1}) &= \mathbb{V}(c + \phi_1 y_{t-1} + \varepsilon_t | \mathcal{F}_{t-1}) \\&= \mathbb{V}(c + \phi_1 y_{t-1} | \mathcal{F}_{t-1}) + \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) \\&= 0 + \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) \\&= \mathbb{V}(\varepsilon_t) \\&= \sigma_\varepsilon^2\end{aligned}$$

Question:

What is the unconditional variance of y_t ?

What is $\mathbb{V}(y_t)$?

Hint: Use Law of Total Variance

Unconditional Variance

$$\mathbb{V}(y_t) = \mathbb{E}(\mathbb{V}(y_t \mid \mathcal{F}_{t-1})) + \mathbb{V}(\mathbb{E}(y_t \mid \mathcal{F}_{t-1}))$$

$$= \mathbb{E}(\sigma_\varepsilon^2) + \mathbb{V}(c + \phi_1 y_{t-1})$$

$$= \sigma_\varepsilon^2 + \phi_1^2 \mathbb{V}(y_{t-1})$$

$$= \sigma_\varepsilon^2 + \phi_1^2 \mathbb{V}(y_t)$$

$$\mathbb{V}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \phi_1^2}$$

A question

What is

$$E(\varepsilon_{t+2} \mid \mathcal{F}_t)?$$

Answer: Apply LIE

$$E(\varepsilon_{t+2} \mid \mathcal{F}_t) = E(E(\varepsilon_{t+2} \mid \mathcal{F}_{t+1}) \mid \mathcal{F}_t) = E(0 \mid \mathcal{F}_t) = 0.$$

Using LIE

In general, for $h \geq 1$ we have

$$E(\varepsilon_{t+h} \mid \mathcal{F}_t) = 0$$

because

$$E(\varepsilon_{t+h} \mid \mathcal{F}_t) = E(E(\varepsilon_{t+h} \mid \mathcal{F}_{t+h-1}) \mid \mathcal{F}_t)$$

$$= E(0 \mid \mathcal{F}_t)$$

$$= 0$$

One-step ahead forecast

A one-step ahead forecast

$$\begin{aligned}\mathbb{E}(y_{t+1}|\mathcal{F}_t) &= c + \phi_1 \mathbb{E}(y_t|\mathcal{F}_t) + \mathbb{E}(\varepsilon_{t+1}|\mathcal{F}_t) \\ &= c + \phi_1 y_t\end{aligned}$$

What about more than one-step ahead?

A **two-step ahead** forecast would be based on

$$\begin{aligned}\mathbb{E}(y_{t+2}|\mathcal{F}_t) &= \mathbb{E}[c + \phi_1 y_{t+1} + \varepsilon_{t+2}|\mathcal{F}_t] \\&= c + \phi_1 \mathbb{E}[y_{t+1}|\mathcal{F}_t] + \mathbb{E}[\varepsilon_{t+2}|\mathcal{F}_t] \\&= c + \phi_1 (c + \phi_1 y_t) + 0 \\&= c(1 + \phi_1) + \phi_1^2 y_t\end{aligned}$$

What about more than one-step ahead?

A three-step ahead forecast

$$\begin{aligned}\mathbb{E}(y_{t+3}|\mathcal{F}_t) &= c + \phi_1\mathbb{E}(y_{t+2}|\mathcal{F}_t) + \mathbb{E}(\varepsilon_{t+3}|\mathcal{F}_t) \\ &= c(1 + \phi_1 + \phi_1^2) + \phi_1^3 y_t.\end{aligned}$$

etc...

Comments on the use of AR models for forecasting

- ▶ Forecasts based on well-specified AR models are consistent (i.e. bias is negligible in large samples).
- ▶ But we need to choose our lag length p carefully!
- ▶ Include enough lags to ensure that there is **no left over serial correlation in the residuals**.
- ▶ Then we can interpret ε_t as ‘news’ (unpredictable market forces).